

FACT FORGE

An Advanced Neural Framework for Real-time Misinformation Analysis

Prepared by:

Yusuf Adel Abbas Amin, Ziad Sameh, Omar Salah

Academic Advisor:

Eng. Mahinda

Institution:

Future University (FUE)
Faculty of Computer Science

Date: May 15, 2026

1 Project Title

Fact Forge: A High-Precision Transformer-Based System for Real-time Misinformation Detection.

2 Problem Definition and Context/Background

In the contemporary digital landscape, the rapid proliferation of misinformation—commonly known as “fake news”—poses a significant threat to global social stability, economic markets, and public trust. Traditional news consumption has shifted from moderated editorial platforms to decentralized social media feeds, where deceptive content can be generated and disseminated at an unprecedented scale.

The core challenge lies in the “evolution of deception.” Modern misinformation is often written with sophisticated linguistic structures that mimic professional journalism, making it difficult for standard rule-based systems or even human readers to distinguish truth from fabrication. There is a critical need for an automated, context-aware system that can analyze the underlying semantic features of a text to provide an objective credibility score without relying on easily manipulated metadata.

3 Project Objectives

The primary objective of **Fact Forge** is to bridge the gap between complex Deep Learning models and end-user accessibility through a robust, working system.

3.1 SystemTasks

- **Neural Classification:** Utilize the DeBERTa-v3 architecture to classify news into “Real” or “Fake” categories with high precision.
- **Confidence Quantification:** Generate a mathematical “Confidence Metric” to inform the user of the model’s certainty.
- **Hardware Acceleration:** Optimize inference using NVIDIA CUDA to ensure the system can handle long-form articles in sub-second timeframes (< 800ms).

3.2 UserInteraction

- **Data Input Terminal:** Users interact with a clean, terminal-style interface where raw text is pasted for neural validation.
- **Visual Feedback System:** The system employs a dynamic UI approach. Upon execution, the interface provides immediate visual cues: a **Green Glow** indicates “Real News,” while a **Red Glow** serves as a high-intensity alert for “Fake News.”
- **Metric Visualization:** A real-time progress bar illustrates the confidence level, providing transparency into the AI’s decision-making process.

4 Methodology

Approach: The project adopts a Transfer Learning approach using the **DeBERTa-v3** (Decoding-enhanced BERT with disentangled attention) model.

Justification: DeBERTa-v3 was chosen over standard BERT due to its "Disentangled Attention" mechanism. By processing content and relative position separately, the model achieves a superior understanding of syntactic nuances and sensationalist intent, which are key indicators of misinformation.

Assumptions:

- The input text is in English (aligned with the WELFake dataset).
- The analysis is purely linguistic; the model evaluates the stylistic "nature" of the writing rather than cross-referencing external databases.

5 Implementation Details

- **Tools:** Python, PyTorch, HuggingFace Transformers, Flask (Backend), React.js (Frontend), and NVIDIA CUDA 12.x.
- **Data:** The **WELFakeDataset** (72,134 records), providing a balanced distribution of labeled news articles.
- **Preprocessing (Nuclear Cleaning):**
 - **Source Neutralization:** Stripping agency names (e.g., Reuters, AP) to prevent source-based bias.
 - **Temporal Stripping:** Removing dates to force the model to focus on the semantic message.
- **Training:** Fine-tuned over 3 Epochs using an AdamW optimizer ($LR = 1 \times 10^{-5}$) and a Dropout rate of 0.5 for high regularization.
- **Testing:** A 20% validation split was utilized to monitor model stability and generalization.

6 Test Results

The implementation demonstrated successful convergence. The final metrics after the third epoch are presented below:

Metric	Result
FinalValidationAccuracy	95.67%
FinalValidationLoss	0.1832
InferenceSpeed	<800ms(onRTX3070)

Table 1: System Performance Metrics

```

===== Epoch 1 / 3 =====
Batch   200 of 3801. Loss: 0.6862
Batch   400 of 3801. Loss: 0.3790
Batch   600 of 3801. Loss: 0.3865
Batch   800 of 3801. Loss: 0.3733
Batch  1000 of 3801. Loss: 0.2978
Batch  1200 of 3801. Loss: 0.1906
Batch  1400 of 3801. Loss: 0.1169
Batch  1600 of 3801. Loss: 0.0696
Batch  1800 of 3801. Loss: 0.1530
Batch  2000 of 3801. Loss: 0.1701
Batch  2200 of 3801. Loss: 0.2762
Batch  2400 of 3801. Loss: 0.2770
Batch  2600 of 3801. Loss: 0.0118
Batch  2800 of 3801. Loss: 0.2762
Batch  3000 of 3801. Loss: 0.1609
Batch  3200 of 3801. Loss: 0.4239
Batch  3400 of 3801. Loss: 0.0185
Batch  3600 of 3801. Loss: 0.0088
Batch  3800 of 3801. Loss: 0.1696

Evaluating Model Stability...
Val Loss: 0.2643 | Accuracy: 0.9373
🌟 New Best Model Saved!

===== Epoch 2 / 3 =====
Batch   200 of 3801. Loss: 0.2070
Batch   400 of 3801. Loss: 0.6119
Batch   600 of 3801. Loss: 0.0178
Batch   800 of 3801. Loss: 0.4291
Batch  1000 of 3801. Loss: 0.0105
Batch  1200 of 3801. Loss: 0.0066
Batch  1400 of 3801. Loss: 0.3136
Batch  1600 of 3801. Loss: 0.4167
Batch  1800 of 3801. Loss: 0.0480
Batch  2000 of 3801. Loss: 0.0495
Batch  2200 of 3801. Loss: 0.0034
Batch  2400 of 3801. Loss: 0.6249
Batch  2600 of 3801. Loss: 0.8458
Batch  2800 of 3801. Loss: 0.2175
Batch  3000 of 3801. Loss: 0.2336
Batch  3200 of 3801. Loss: 0.1421
Batch  3400 of 3801. Loss: 0.5133
Batch  3600 of 3801. Loss: 0.0179
Batch  3800 of 3801. Loss: 0.0025

Evaluating Model Stability...
Val Loss: 0.2206 | Accuracy: 0.9537
🌟 New Best Model Saved!

```

```

===== Epoch 3 / 3 =====
Batch   200 of 3801. Loss: 0.0492
Batch   400 of 3801. Loss: 0.0137
Batch   600 of 3801. Loss: 0.0969
Batch   800 of 3801. Loss: 0.1801
Batch  1000 of 3801. Loss: 0.3333
Batch  1200 of 3801. Loss: 0.3720
Batch  1400 of 3801. Loss: 0.1278
Batch  1600 of 3801. Loss: 0.0897
Batch  1800 of 3801. Loss: 0.4064
Batch  2000 of 3801. Loss: 0.0208
Batch  2200 of 3801. Loss: 0.2812
Batch  2400 of 3801. Loss: 0.0520
Batch  2600 of 3801. Loss: 0.8511
Batch  2800 of 3801. Loss: 0.0031
Batch  3000 of 3801. Loss: 0.0330
Batch  3200 of 3801. Loss: 0.1643
Batch  3400 of 3801. Loss: 0.2960
Batch  3600 of 3801. Loss: 0.1910
Batch  3800 of 3801. Loss: 0.0298

Evaluating Model Stability...
Val Loss: 0.1832 | Accuracy: 0.9567
🌟 New Best Model Saved!

```

7 Discussion

Evaluation: The 95.67% accuracy confirms that DeBERTa-v3 is exceptionally suited for this domain. The low validation loss indicates a high degree of "Semantic Certainty."

Strengths: The system is resilient against bias leakage due to the strict preprocessing protocol and offers a professional, hardware-accelerated user experience.

Weaknesses: Currently limited to English-language text. It does not account for multi-modal deception (e.g., edited images or deepfake video).

Improvements: Future versions will integrate **AraBERT** for Arabic support and **Explainable AI (XAI)** tools to highlight specific deceptive phrases within the UI.